CHAMP: Dynamic Harmonisation and Processing of Diverse CRNS Datasets

A Metadata-Driven Approach

Merlin Schiel[1], Till Francke[1], Jakob Terschlüsen[1], Sascha E. Oswald[1]

[1]Universität Potsdam, Institute of Environmental Science and Geography

1. Scaling CRNS Networks

Challenges

Data Format Diversity

Multiple manufacturers, loggers, and firmware versions producing incompatible raw outputs.

Lifecycle Changes

Sensor swaps, relocations, and maintenance gaps creating temporal inconsistencies.

Multi-Source Integration

Merging auxiliary sensors at varying temporal resolutions into unified time series.

Solution

Data Harmonisation

Metadata-driven framework that automatically ingests, aligns, and processes diverse CRNS data streams

2. Concept of Sensor "Phases"

Long-term monitoring stations are dynamic systems. Hardware swaps, firmware upgrades, relocations, and environmental shifts alter data characteristics over time:

- **Why "Phases"?** Every significant change in a sensor's lifecycle defines a new *phase*: a distinct operational period with its own data format, calibration, and site properties.
- **Simple Management:** Phases are defined by timestamps in a spreadsheet. Each phase receives its own metadata configuration.
- **Key Advantages:**
  - **Continuous Time Series:** Produces an unbroken, multi-year record per station by stitching phases together.
  - **Zero Code Changes:** Adding a new logger format or site parameter requires only a new phase entry.
  - **Full Auditability:** Every sensor change is transparently documented and traceable.

3. Sensor Metadata

- **YAML-Based Configuration:** Raw data structure, sensor properties, and environmental parameters are stored in human-readable YAML files.
- **Native Neptoon Compatibility:** The YAML schema directly maps to the `neptoon` processing library, enabling seamless soil moisture processing.

```yaml
# Sensor Phase Metadata (YAML)
sensor_info:
  name: sensor_01
  location: test_site
  identifier: '01'
  install_date: 2026-01-01
  latitude: 44.4949
  longitude: 11.3426
  elevation: 54
  time_zone: 1
  site_cutoff_rigidity: 5.12
  avg_lattice_water: 0.0074
  avg_soil_organic_carbon: 0.0285
  avg_dry_soil_bulk_density: 1.3949
  above_ground_biomass: 20.0
  N0: 3000

raw_data_parse_options:
  separator: ","
  skip_lines: 0
  column_names: [RecordNum, Date_Time, Press, Temp, RH, Batt,
    N1, N1ET]
```

4. Modular Pre-Processing

Instead of modifying source code for each data issue, the YAML format enables modular, declarative pre-processing controls:

- **Auxiliary Sensor Merging:** Automatically integrates data from secondary detectors and soil probes into the primary data stream.
- **Property-Based Data Matching:** Fill your metadata with custom properties (e.g., location), by which data can be flexibly organized.
- **Timestamp Corrections:** Resolves clock drifts, timezone offsets, and irregular intervals with configurable alignment rules.
- **Extensible Architecture:** New pre-processing steps can be added as YAML rules without altering the core pipeline code.

Configuration Over Code

Declarative YAML rules handle diverse sensor issues and pre-processing tasks without altering processing scripts.

5. Soil Moisture Processing with neptoon

All scientific processing, quality assessment, atmospheric corrections, calibration, and soil moisture conversion is delegated to the `neptoon` processing engine:

- **Direct Pipeline Delegation:** Harmonised data and YAML configurations are passed directly to `neptoon`'s processing workflow.
- **Integrated Calibration:** In-situ soil sampling campaigns are processed automatically and calibration results (N_0) are written back into sensor metadata.
- **Phase-Aware Processing:** Each phase can use independent calibration and correction parameters, which are then merged into a single continuous time series.

Neptoon Handles the Science

All corrections, calibration, and soil moisture conversion are fully delegated, ensuring accuracy and reproducibility.

8. Summary

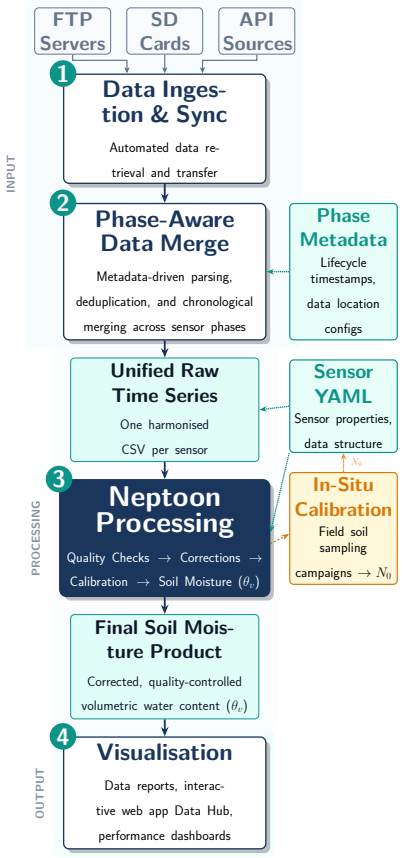
1. **Scalable Pipeline:** Eliminates manual data handling across expanding CRNS networks.
2. **Tracking Lifecycle Changes:** Seamlessly handles sensor swaps, firmware upgrades, and calibration shifts through metadata alone.
3. **Neptoon Integration:** Fully utilizes `neptoon` for reproducible data processing and soil moisture estimation.
4. **User-Friendly:** Modular YAML configuration and an interactive Data Hub simplify network operations for diverse user groups.

Open-Source on GitHub

`https://github.com/merlinschiel/data_champ`

Complete codebase, sample configurations & documentation.

6. End-to-End Architecture



7. Interactive Data Hub

- **Web-Based UI:** Bridges automated backend processing with intuitive, streamlit-based data exploration
- **Sensor Overview:** Overview of your sensor data along with network map.
- **Hardware Monitoring:** Continuous monitoring dashboard of the most recent raw data coming from your sensors
- **Quality Control & Export:** Plotly-based visual data validation and export of clean datasets.

